

Disjoint-union bounds for the continuous theta saturation parameter

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190
xiaojianzeng@foxmail.com

Abstract

For the continuous semidefinite theta saturation parameter μ , we prove lower and upper bounds on a disjoint union in terms of its parts. The bounds coincide when every part attains the known product inequality $\vartheta(G)\mu(G) \geq |V(G)|$. An explicit coupling of component matrices gives the upper bound. The component hypothesis cannot simply be omitted: $\mu(P_3 \sqcup K_1) = \sqrt{3} < 7/4$. Join additivity and the union bound imply that any least-order counterexample to $\mu(G) \leq \vartheta(\overline{G})$ must be connected with connected complement. Exact certificates and exhaustive coverage establish this inequality for all graphs on at most six vertices. The universal inequality remains unresolved by these results; no broad priority claim is made.

1 Definitions and scope

All graphs are finite and simple. Unless stated otherwise, they have positive order. Write $J_m = \mathbf{1}_m \mathbf{1}_m^\top$, $P_m = J_m/m$, and $\langle A, B \rangle = \text{Tr}(A^\top B)$. All matrix variables are real symmetric; $\preceq$ denotes the positive-semidefinite order. For a graph G on n vertices, use precisely the continuous SDP in Carli Silva, Coutinho, Oliveira and Tunçel [1, Eq. (14)]:

$$\mu(G) = \min\{\text{Tr } X : 0 \preceq X \preceq I_n, X_{uv} = 0 \ (uv \in E(G)), \langle J_n, X \rangle = n\}, \quad (1)$$

$$\vartheta(G) = \max\{\langle J_n, Z \rangle : Z \succeq 0, \text{Tr } Z = 1, Z_{uv} = 0 \ (uv \in E(G))\}. \quad (2)$$

In particular, μ is not rounded to an integer, and there is no entrywise nonnegativity constraint. We extend both parameters to the graph with no vertices by the separate convention $\mu(\emptyset) = \vartheta(\emptyset) = 0$. This is not an application of the trace-one program to a zero-order matrix.

The motivating question is whether

$$\mu(G) \leq \vartheta(\overline{G}) \quad (3)$$

holds for every graph. It is posed in [1, Section 2.2 and question (iii)]. The same source proves $\vartheta(G)\mu(G) \geq n$ and equality when (2) has a constant-diagonal optimum. It also gives a feasible matrix with trace $5/3$ for $K_2 \sqcup K_1$. Our main statement supplies componentwise bounds, an attaining construction under a sufficient hypothesis, and an exact boundary to the unconditional equality assertion.

The comparison with prior work is limited to the supplied version arXiv:2609.06312v1. Its Proposition 15 explicitly points to a formula for $\vartheta_k(K_n \sqcup K_1)$ in Oliveira's thesis [2]. Its Section 2.2 also discusses the integer saturation predecessor in Sinjorgo and Sotirov [3]. Those two primary works were not inspected directly here; their bibliographic records and these relationships are reported from [1]. Equivalent statements in the wider literature have not been ruled out. Thus the construction below is compared with the inspected source, without asserting worldwide novelty or a current worldwide status for the question.

2 Bounds on disjoint unions

Theorem 1. *Let $k \geq 1$ and let $G_1, \dots, G_k$ be graphs of positive orders $n_1, \dots, n_k$. Put $N = \sum_i n_i$ and $D = \bigsqcup_{i=1}^k G_i$. Then*

$$\frac{1}{N} \sum_{i=1}^k \frac{n_i^2}{\vartheta(G_i)} \leq \mu(D) \leq \frac{1}{N} \sum_{i=1}^k n_i \mu(G_i). \quad (4)$$

If $\vartheta(G_i)\mu(G_i) = n_i$ for every i , equality holds in both bounds.

Proof. If X is feasible for (1) on m vertices, then $\mathbf{1}_m^\top (I_m - X) \mathbf{1}_m = 0$. Since $I_m - X \succeq 0$, this implies $X \mathbf{1}_m = \mathbf{1}_m$. Symmetry and the spectral constraints therefore give

$$X = P_m + R, \quad R \mathbf{1}_m = 0, \quad 0 \preceq R \preceq I_m - P_m. \quad (5)$$

Conversely, row sums one give the required equality $\langle J_m, X \rangle = m$. The feasible sets in (1) and (2) are nonempty and compact: I_m and I_m/m , respectively, are feasible, and the trace or spectral constraints bound them. Thus optima exist, and $\vartheta(G_i) \geq 1$.

For any positive-semidefinite Z vanishing on the edges of G_i , scaling in (2) gives

$$\langle J_{n_i}, Z \rangle \leq \vartheta(G_i) \operatorname{Tr} Z. \quad (6)$$

The zero matrix is included, since a nonzero positive-semidefinite matrix has positive trace. Now let X be feasible for D . Its principal block X_{ii} satisfies $X_{ii} \succeq J_{n_i}/N$ by (5). Consequently

$$\operatorname{Tr} X_{ii} \geq \frac{\langle J_{n_i}, X_{ii} \rangle}{\vartheta(G_i)} \geq \frac{n_i^2}{N \vartheta(G_i)}.$$

Summing over i proves the lower bound.

For the upper bound, choose optimal component matrices X_i and set

$$X = P_N + \operatorname{diag}_i \left[\frac{n_i}{N} (X_i - P_{n_i}) \right]. \quad (7)$$

Each residual block is positive semidefinite, has norm at most n_i/N , and acts on the vectors supported in part i whose coordinates sum to zero. These spaces are mutually orthogonal and orthogonal to the global constant vector. Hence $0 \preceq X \preceq I_N$ and $X \mathbf{1}_N = \mathbf{1}_N$. The diagonal block in part i is exactly $(n_i/N)X_i$; cross-part entries are $1/N$ and there are no cross-part edges. Thus X is feasible. Its trace is

$$1 + \sum_i \frac{n_i}{N} (\mu(G_i) - 1) = \frac{1}{N} \sum_i n_i \mu(G_i),$$

as required. Under the displayed component hypothesis the lower and upper expressions coincide. $\square$

The lower-bound argument also works for a vertex partition into induced subgraphs; the upper construction uses the absence of edges between parts. The theorem includes $k = 1$, singleton parts, repeated parts, and unequal part sizes. The equality hypothesis is sufficient; no necessity is asserted.

For comparison, the product bound applied to D gives $\mu(D) \geq N / \sum_i \vartheta(G_i)$, using the theta union identity proved below. Cauchy–Schwarz gives

$$\frac{1}{N} \sum_i \frac{n_i^2}{\vartheta(G_i)} \geq \frac{N}{\sum_i \vartheta(G_i)},$$

with equality precisely when all $n_i/\vartheta(G_i)$ coincide. For $K_2 \sqcup K_1$ the two lower bounds are $5/3$ and $3/2$.

Lemma 2 (Constant-diagonal case; [1, Theorem 6]). *If a graph G of order n has a constant-diagonal optimum for $\vartheta(G)$, then $\vartheta(G)\mu(G) = n$. In particular this holds for vertex-transitive graphs.*

Proof. Put $q = \vartheta(G)$ and let Z be such an optimum. Its diagonal is $1/n$. For every unit vector h , the matrix $n \operatorname{diag}(h)Z \operatorname{diag}(h)$ is feasible in (2). Hence $nh^\top Zh \leq q$ and $\lambda_{\max}(Z) \leq q/n$. Therefore $X = (n/q)Z$ is feasible in (1), with trace n/q . Conversely, normalizing any μ -feasible X to trace one gives $q \geq n/\operatorname{Tr} X$, which is the matching lower bound. Averaging a theta optimum over the automorphism group of a vertex-transitive graph makes its diagonal constant. $\square$

Corollary 3. *For integers $r \geq 0$ and positive integers $n_1, \dots, n_k$,*

$$\mu(C_5 \sqcup rK_1) = \frac{5\sqrt{5} + r}{5 + r}, \quad \mu\left(\bigsqcup_i K_{n_i}\right) = \frac{\sum_i n_i^2}{\sum_i n_i}.$$

Proof. Directly, $\vartheta(K_m) = 1$ and $\mu(K_m) = m$. Here is an exact certificate for the other required value. Let A be the adjacency matrix of C_5 , $B = J_5 - I_5 - A$, and $\rho = (\sqrt{5} - 1)/2$. Then

$$X_5 = \frac{I_5 + \rho B}{\sqrt{5}}, \quad Y_5 = \frac{I_5 + \rho A}{5} \quad (8)$$

are respectively $\mu(C_5)$ -feasible and $\vartheta(\overline{C_5})$ -feasible. Their eigenvalues are $1, \rho, \rho, 0, 0$ and $1/\sqrt{5}, (5 - \sqrt{5})/10, (5 - \sqrt{5})/10, 0, 0$. Also $X_5/\sqrt{5}$ is theta-feasible for C_5 with objective $\sqrt{5}$. The matrix

$$S = \sqrt{5}I_5 + \frac{5 - \sqrt{5}}{2}A - J_5$$

has eigenvalues 0 with multiplicity three and $(5\sqrt{5} - 5)/2$ with multiplicity two. For any theta-feasible Z , $0 \leq \langle S, Z \rangle = \sqrt{5} - \langle J_5, Z \rangle$. Thus $\vartheta(C_5) = \sqrt{5}$, and the product bound together with X_5 gives $\mu(C_5) = \sqrt{5}$. Apply Theorem 1. $\square$

3 An exact strictness example

Proposition 4. *The upper bound in (4) need not be an equality. In fact,*

$$\mu(P_3) = 2, \quad \mu(P_3 \sqcup K_1) = \sqrt{3} < \frac{3\mu(P_3) + \mu(K_1)}{4} = \frac{7}{4}.$$

Proof. Label $P_3 \sqcup K_1$ by 0, 1, 2, 3, with edges 01 and 12. Average any feasible matrix under the interchange of leaves 0 and 2. Writing the result as $X = I_4 - L$, the row sums and edge zeros force L to be the signed Laplacian with weight a on 02, weight b on each of 03, 23, and weight c on 13. The weights are arbitrary real numbers in the lower-bound argument. Feasibility means $0 \preceq L \preceq I_4$.

On the leaf difference vector, L has eigenvalue $2a + b \leq 1$. On the invariant subspace of vectors with equal leaf coordinates, the quotient matrix in coordinates (leaf, center, isolate) is

$$\begin{pmatrix} b & 0 & -b \\ 0 & c & -c \\ -2b & -c & 2b + c \end{pmatrix},$$

whose characteristic polynomial is $z[z^2 - (3b + 2c)z + 4bc]$. Although these coordinates are not orthonormal, this is a restriction of a symmetric operator. Its two residual eigenvalues are $(u \pm \sqrt{v})/2$, where

$$u = 3b + 2c \geq 0, \quad v = 9b^2 - 4bc + 4c^2, \quad v - \frac{u^2}{3} = \frac{2}{3}(3b - 2c)^2 \geq 0.$$

The larger eigenvalue is at most one, so $u(1 + 1/\sqrt{3})/2 \leq 1$ and $u \leq 3 - \sqrt{3}$. It follows that

$$\text{Tr } X = 4 - 2a - 4b - 2c \geq 3 - 3b - 2c = 3 - u \geq \sqrt{3}.$$

Equality is attained by

$$a = \frac{3 + \sqrt{3}}{12}, \quad b = \frac{3 - \sqrt{3}}{6}, \quad c = \frac{3 - \sqrt{3}}{4}.$$

The associated X has row sums one, the required zeros, and spectrum $1, \sqrt{3} - 1, 0, 0$. It is feasible and has trace $\sqrt{3}$.

For P_3 alone, the center is a singleton block equal to one, while the two-leaf block is positive semidefinite with row sums one, hence trace at least one. The leaf block $J_2/2$ attains this bound. Finally, $\mu(K_1) = 1$ and $3 < 49/16$ prove the strict comparison. $\square$

This refutes only the unconditional component equality, not (3).

4 Joins and a reduction for counterexamples

The join $\bigvee_i G_i$ adds every edge between distinct parts.

Proposition 5. *For positive-order graphs,*

$$\mu\left(\bigvee_i G_i\right) = \sum_i \mu(G_i).$$

The class satisfying (3) is closed under disjoint unions and joins. Consequently, any counterexample with the least positive number of vertices is connected and has connected complement.

Proof. On a join, the edge constraints force every μ -feasible matrix to be block diagonal. Row sums one and both spectral bounds hold blockwise. Thus minimizing the trace gives the stated sum.

We include proofs of the standard theta identities needed here:

$$\vartheta\left(\bigvee_i H_i\right) = \max_i \vartheta(H_i), \quad \vartheta\left(\bigsqcup_i H_i\right) = \sum_i \vartheta(H_i). \quad (9)$$

For the join, a theta-feasible matrix is block diagonal. Each block objective is at most its trace times its component theta value by (6). Concentrating trace one in an optimizing component attains the maximum.

For the union, write a feasible Y as a Gram matrix, and put $t_i = \text{Tr } Y_{ii}$ and $a_i = \langle J, Y_{ii} \rangle$. The triangle inequality for the sums of Gram vectors and Cauchy–Schwarz yield

$$\langle J, Y \rangle \leq \left(\sum_i \sqrt{a_i} \right)^2 \leq \left(\sum_i \sqrt{\vartheta(H_i) t_i} \right)^2 \leq \sum_i \vartheta(H_i).$$

For attainment, let Z_i be trace-one optima, set $q_i = \vartheta(H_i)$, $T = \sum_i q_i$, and $v_i = Z_i \mathbf{1}$. Concatenate the v_i to form v . Matrix Cauchy–Schwarz gives $v_i v_i^\top \preceq q_i Z_i$, so

$$Y = \frac{vv^\top}{T} + \text{diag}_i \left[\frac{q_i Z_i - v_i v_i^\top}{T} \right]$$

is positive semidefinite. Its diagonal blocks are $q_i Z_i / T$, its trace is one, and its objective is T : each residual block has objective zero and $\mathbf{1}^\top v = T$. This proves (9).

If each G_i satisfies (3), the union upper bound gives

$$\mu \left(\bigsqcup_i G_i \right) \leq \sum_i \frac{n_i}{N} \vartheta(\overline{G_i}) \leq \max_i \vartheta(\overline{G_i}) = \vartheta \left(\overline{\bigsqcup_i G_i} \right).$$

For a join, both parameters in (3) add. A disconnected least-order counterexample would have smaller components all satisfying the inequality, contradicting closure. If its complement were disconnected, the graph would be a join of smaller induced graphs, giving the same contradiction. The null graph is already covered by convention. $\square$

These identities are supporting tools, not separate claims of novelty. The connectivity condition is necessary, with no asserted converse.

5 Exact verification through six vertices

Proposition 6. *Every graph with at most six vertices satisfies (3), with the null-graph convention stated above.*

Proof by exact finite certificates. A proper coloring with q nonempty color classes gives a μ -feasible matrix consisting of blocks $J_C/|C|$, with zero cross-class entries. It is an orthogonal projection of trace q . A clique C of size q gives the complementary-theta certificate $\mathbf{1}_C \mathbf{1}_C^\top / q$, with trace one and objective q . Thus a matching coloring and clique prove (3).

Among the 208 positive-order graph-atlas representatives through order six, 203 have such stored witnesses. The five exceptions are C_5 and the four six-vertex graphs in Table 1. The C_5 case follows from (8). Each six-vertex graph contains the cycle with edges 01, 12, 23, 34, 40 on vertices $0, \dots, 4$. The table specifies the neighbors of the sixth vertex and exact certificate bounds.

Atlas index	Neighbors of vertex 5	μ upper bound	$\vartheta(\overline{G})$ lower bound
76	$\emptyset$	51/25	$\sqrt{5}$
104	$\{1\}$	272/125	$\sqrt{5}$
129	$\{0, 3\}$	279/125	$\sqrt{5}$
187	$\{0, 1, 2, 3, 4\}$	$1 + \sqrt{5}$	$1 + \sqrt{5}$

Table 1: Exact certificate directions. The first three upper bounds are not asserted optimal.

For the first three rows, put $Y = \text{diag}(Y_5, 0)$ and $X = I_6 - \sum_{uv \in E(\overline{G})} w_{uv}(e_u - e_v)(e_u - e_v)^\top$. All weights are positive and are given in Appendix A. Consequently $I_6 - X \succeq 0$, the row sums of X are one, and its entries vanish on the graph edges. Its six leading principal minors are strictly positive by exact rational evaluation (the full values are in the supplement), so $X \succ 0$ by Sylvester's criterion. Direct traces give the table. Each rational upper bound is below $\sqrt{5}$, as can be checked by squaring positive rational numbers. For the wheel in the last row, take $X = \text{diag}(X_5, 1)$ and use the theta union construction in the proof of Proposition 5 on $\overline{C_5} \sqcup K_1$. This gives trace and objective $1 + \sqrt{5}$.

For completeness of the finite domain, expand each atlas representative under every permutation of its labeled vertex set. Encode edges using bit positions indexed by the unordered pairs. For each $n = 1, \dots, 6$, the resulting set of masks equals $\{0, \dots, 2^{\binom{n}{2}} - 1\}$. The exact counts are

n	1	2	3	4	5	6
Representatives	1	2	4	11	34	156
Labeled graphs	1	2	8	64	1024	32768

Thus every one of the 33,867 positive-order labeled graphs in the stated range is covered. The supplementary JSON files retain all 203 coloring and clique witnesses, the exceptional matrices, and the coverage counts. The verifier checks their supports, linear constraints, exact objectives, and signs, and recomputes the mask-set equality. This proves the finite statement only. $\square$

Reproducibility and assistance. The supplement contains the exact data and a verifier using Python, NetworkX and SymPy; its README gives the execution command. Rational and quadratic-field arithmetic, rather than floating-point feasibility tolerances, determine the certificate signs. In a separate verification pass, every nonempty principal minor of X , $I - X$ and Y was recomputed for the four six-vertex certificates (756 minors in total); the C_5 and strictness certificates were also checked. Coverage was independently recomputed by permutation expansion, in addition to the discovery pass’s exhaustive isomorphism matching. The verifier is an inspectable computation, not a formal proof-assistant formalization.

OpenAI Codex assisted with proof development, exploratory numerical searches, exact verification code, a separate mathematical review pass, and manuscript preparation. The separate review used a fresh model context and does not constitute external peer review. SciPy SLSQP outputs guided rational certificate selection; none is used as decisive evidence, including an unsuccessful numerical solve. Zeng’s spectral-graph certification study [4] is cited only as adjacent context for exact matrix certificates and finite graph enumeration; it supplies no theorem used here.

Remaining scope. Theorem 1, Proposition 4, and Proposition 5 concern unrestricted component orders, whereas Proposition 6 has the explicit upper limit of six vertices. No certificate for all larger connected graphs with connected complement is supplied. The universal inequality (3) remains unresolved by this note, and the literature advisories in Section 1 apply to any assessment of originality.

A Rational Laplacian data

In Table 2, uv denotes the unordered pair $\{u, v\}$. Each listed weight applies to every pair in its row. These rows partition the nonedges of the corresponding graph in Table 1. All six

Atlas index	Pairs	Weight
76	02, 03, 13, 14, 24	23/100
	05, 15, 25, 35, 45	83/500
104	02	9/50
	03, 24	237/1000
	05, 25	127/500
	13, 14	257/1000
	35, 45	59/500
129	02, 13	27/125
	03	163/500
	14, 15, 24, 25	209/1000
	45	29/100

Table 2: Exact positive weights defining the three rational certificates.

leading principal minors of each X are positive. For compactness, their numerators in lowest

terms, ordered by leading block size, are

76	187	34969	254133	10336511	23819457	57121
104	329	79947	18429363	310877393	296786591	734421
129	229	41907	5534469	1554363	10817613	37053.

The corresponding positive denominators and serialized matrices are retained in `six_vertex_certificates.json` in the supplement. All these determinants can also be reconstructed directly from Table 2.

References

- [1] M. K. Carli Silva, G. Coutinho, T. Oliveira, and L. Tunçel, *On some k -fold generalizations of Lovász theta and their sandwich theorems*, arXiv:2609.06312v1 (2026). <https://arxiv.org/abs/2609.06312v1>.
- [2] T. Oliveira, *The Maximum k -Colorable Subgraph Problem*, master’s thesis, Institute of Mathematics, Statistics, and Computer Science, July 2024. <https://doi.org/10.11606/D.45.2024.tde-25082025-111037>. Bibliographic record and predecessor pointer reported from [1].
- [3] L. Sinjorgo and R. Sotirov, *On the generalized ϑ -number and related problems for highly symmetric graphs*, SIAM Journal on Optimization **32**(2) (2022), 1344–1378. <https://doi.org/10.1137/21M1414620>. Bibliographic record reported from [1].
- [4] Zijian Zeng, *Exact Finite Certification of The Sparse-Complement Two-Eigenvalue Conjecture at Order Nine*, SSRN (2026). <https://doi.org/10.2139/ssrn.7379578>.